Correlation

Previously we are used to study the characteristics of only one variable like marks, weights, prices, ages, Salaries This type of analysics is called Univariate Analysis. If there exists some relation_ship between two variables then the statistical analysis of such data is called Bivariate Analysis. We may

Definition:

Correlation is a statistical Analysis which measures and analysis the degree or extent to which two variables fluctuate with reference to each other. The correlation express the relation-ship or interdepen of two set of variables upon each other. one variable may called the subject (independent) and the other (relative) is called dependent.

Types of Correlation:

Correlation is classified into many (ways) types

1. Positive and Nagative
2. Simple and Multiple
3. Partial and Total
4. Linear and Non-linear.

1. Positive and Nagative Correlation:

positive and Nagative Correlation depend upon the direction of change of the variables.

If two variables tend to move together in the same direction. i.e., an increase value of the one variable is accompained by an increase in the value of the other variable; or a decrease in the value of one variable is accompained by a decrease in the value of the other variable, then the correlation is called positive or direct correlation.

2. Simple and Multiple Correlation:

When we study only two variables, the relationship is described as simple correlation.

Example are quantity of money and price level.
demand and price etc.

But in a multiple correlation we study more than two variables simultaneously.

Example is the relationship of price, demand and supply of a commodity.

Partial and Total Correlation:

The study of two variables excluding some other variables is called partial correlation.

For example, we study price and demand, eliminating the supply side. In total correlation, all facts are taken into the account.

4. Linear and Non-Linear Correlation:

If the ratio of change between two variables is uniform, then there will be linear correlation between them. Non-linear Correlation. the amount of change in one variable does not bear a constant ratio of the amount of change in the other variables.

Mathematical Methods are.

(a) Karl Pearson's Coefficient of Correlation

(b) Spearman's Rank coefficient of correlation.

Covariance:

Def: Let X and Y be the two random variables with means $\bar{X}$ and $\bar{Y}$ respectively.
Then the covariance between two variables X and Y is defined by the relation

$$Cov(X,Y) = E[(X-\bar{X})(Y-\bar{Y})].$$

Coefficient of Correlation:

Correlation is a statistical technique used for analysing the behaviour of two or more variables. Its analysis deals with association, between two or more variables.

Karl Pearson's coefficient Coefficient:

Karl pearson's coefficient of correlation or correlation coefficient r between the variables X and Y is defined by the formula

$$r = \frac{\text{Covariance of } X \text{ and } Y.}{(\text{S.D of } X)\ (\text{S.D. of } Y).}$$

$$= \frac{\text{cov}(X,Y)}{\sigma_x \cdot \sigma_y} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n \cdot \sigma_x \sigma_y}.$$

σ_x S.D of x $\quad$ σ_y S.D of Y.

We know that $\sigma_x = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(x-\bar{x})^2}$ and

$$\sigma_y = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y-\bar{y})^2}$$

Another form of the formula for the correlation coefficient r.

$$r = \frac{1}{n}\sum_{i=1}^{n} \frac{x_i y_i - \bar{x}\bar{y}}{\sqrt{\frac{1}{n}\sum_{i=1}^{n} x_i^2 - (\bar{x})^2} \times \sqrt{\frac{1}{n}\sum_{i=1}^{n} y_i^2 - (\bar{y})^2}}$$

Properties of Correlation Coefficient:

Property 1. The maximum value of rank correlation is 1.

The coefficient of correlation lies between -1 and 1.

Symbolically $-1 \le r \le 1$ or $|r| \le 1$.

if $r = 1$ correlation is perfect and positive

if $r = -1$ " " " $-ve$

if $r = 0$ there is no relationship between the variables

Prob: Calculate coefficient of correlation from the following data.

X	12	9	8	10	11	13	7
Y	14	8	6	9	11	12	3.

Sol: In both series items are in small numbers

So there is no need to take deviations

We use the formula $r = \dfrac{\text{Cov. of } (X, Y)}{\sigma_x \, \sigma_y}$

Computation of coefficient of correlation.

X	Y	x^2	y^2	XY
12	14	144	196	168
9	8	81	64	72
8	6	64	36	48
10	9	100	81	90
11	11	121	121	121
13	12	169	144	156
7	3	49	9	21
$\Sigma X = 70$	$\Sigma Y = 63$	$\Sigma x^2 = 728$	$\Sigma y^2 = 651$	$\Sigma XY = 676$

i.e., $r = \dfrac{\Sigma xy}{\Sigma x^2 \cdot \Sigma y^2}$ or $\dfrac{(\Sigma xy \times N) - (\Sigma x \times \Sigma y)}{\sqrt{(\Sigma x^2 \times N - (\Sigma x)^2)(\Sigma y^2 \times N - (\Sigma y)^2)}}$

Here N = No. of obs. = 7.

$$\therefore r = \frac{(676 \times 7) - (70 \times 63)}{\sqrt{728 \times 7 - (70)^2}\sqrt{651 \times 7 - (63)^2}}$$

$$= \frac{322}{339.48} = 0.95.$$

Prob: Find the coeff of correlation between the two variables.

x	50	50	55	60	65	65	6560	60	60
y	11	13	14	16	16	15	1514	13	13

x	y.	$X = x - \bar{x}$	$Y = y - \bar{y}$	X^2	Y^2	XY
50	11	−9	−3	81	9	27
50	13	−9	−1	81	1	9
55	14	−4	0	16	0	0
60	16	1	2	01	4	2
65	16	6	2	36	4	12
65	15	6	1	36	1	6
65	15	6	1	36	1	6
60	14	1	0	1	0	0
60	13	1	−1	1	1	−1
60	13.	1	−1	1	1	−1
Σx 590	$\Sigma y = 140$	0	0	Σx^2 290	Σy^2 22	60

$\bar{x} = \frac{590}{n=10} = 59$; $\bar{y} = \frac{140}{10} = 14$.

Correlation coefficient $r = \dfrac{\Sigma xy - \frac{\Sigma x \Sigma y}{N}}{\sqrt{\Sigma x^2 - \frac{(\Sigma x)^2}{N}}\sqrt{\Sigma y^2 - \frac{(\Sigma y)^2}{N}}}$

$= \dfrac{60 - 0}{\sqrt{290 - 0}\sqrt{22}}$ $= 0.7512$.

or $r = \dfrac{\Sigma xy}{\sqrt{\Sigma x^2 \times \Sigma y^2}} = \dfrac{60}{\sqrt{290 \times 22}}$.

When Deviations are taken from an Assumed mean,

When actual mean is not a whole number, but a fraction or when the series is large, the calculation by direct method will involve a lot of time.

To avoid such tedious calculations, we can use the assumed mean method.

Prob: Calculate Karl pearson's correlation coefficient for the following paired data.

X	28	41	40	38	35	33	40	32	36 33
Y	23	34	33	34	30	26	28	31	36 38.

Here $N=10$ $\bar{x}=35$ taken $\bar{y}=31$

x	$X=x-35$, $X=x-\bar{x}$ Deviation from Assumed mean.	X^2	Y	$y=y-31$ $y=y-\bar{y}$	Square of deviations y^2	Product of deviations XY
28	−7	49	23	−8	64	56
41	+6	36	34	+3	9	18
40	+5	25	33	2	4	10
38	+3	9	34	3	9	9
35	0	0	30	−1	1	0
33	−2	4	26	−5	25	10
40	+5	25	28	−3	9	−15
32	−3	9	31	0	0	0
36	+1	1	36	5	25	5
33	−2	4	38	7	49	−4

$\Sigma x=355$ $\Sigma X=6$ $\Sigma X^2=162$ $\Sigma Y=313$ $\Sigma Y=3$ $\Sigma y^2=195$ $\Sigma xy=79$

$$r = \frac{\Sigma xy - \frac{\Sigma x \cdot \Sigma y}{N}}{\sqrt{\left[\Sigma x^2 - \frac{(\Sigma x)^2}{N}\right] \times \left[\Sigma y^2 - \frac{(\Sigma y)^2}{N}\right]}}$$

$$= \frac{79 - \frac{6 \times 3}{10}}{\sqrt{162 - \frac{6^2}{10}} \cdot \sqrt{195 - \frac{3^2}{10}}} = 0.45$$

$$r = \frac{\Sigma xy \times N - \Sigma x \cdot \Sigma y}{\sqrt{\Sigma x^2 \times N - (\Sigma x)^2 \times \Sigma y^2 \cdot N - (\Sigma y)^2}}$$

Rank Correlation Coefficient:

A British Pychologist Charles Adward Spearman find out the method of finding coefficient of correlation by ranks. This method is based on rank and is usefull in dealing with qualitative characteristics such as morality, charcter and intelligence. It cannot be measured quantitatively as in the case of pearsons coefficient correlation. It is based on the ranks given to the observations. Rank correlation is applicable only to the individual observations.

The formula for spearman's rank correlation is given by. $$\rho = 1 - \frac{6\Sigma d^2}{n(n^2-1)}.$$

Where $\rho \rightarrow$ Rank of correlation.

$D^2 \rightarrow$ Sum of the squares of two ranks

$N \rightarrow$ Numbes of paired observations.

Properties of Rank correlation:

1. The value of ρ lies between +1 and −1.
2. If $\rho = 1$ there is complete agreement in the order if the ranks and directions of the rank sa
3. If $\rho = -1$ then there is complete disagreement the ranks they are in opposite directions

Prob: A random sample of 5 college students is selected and their grades in Mathematics and Statistics are found to be.

Maths	85	60	73	40	90
Stats	93	75	65	50	80.

Marks in Maths (x)	Marks in Stats	Ranks X.	Ranks Y	Rank difference D = x − y	D^2
85	93	2	1	1	1
60	75	4	3	1	1
73	65	3	4	−1	1
40	50	5	5	0	0
90	80.	1	2	−1	1
					$\Sigma d^2 = 4$

$N = 5$

∴ Rank relation $\rho = 1 - \frac{6\Sigma d^2}{n(n^2-1)}$

$= 1 - \frac{6 \times 4}{5.24} = \underline{\underline{0.8}}$

Prob: Following are the ranks obtained by 10 students in two subjects, Statistics and Mathematics. To what extent the knowledge of the students in two subjects is related?

Statistics	1	2	3	4	5	6	7	8	9	10
Mathematics	2	4	1	5	3	9	7	10	6	8

Sol:

Rank in statistics x	Rank in Mathematics y	$D = x - y$	D^2
1	2	-1	1
2	4	-2	4
3	1	2	4
4	5	-1	1
5	3	2	4
6	9	3	9
7	7	0	0
8	10	-2	4
9	6	3	9
10	8	2	4
		Σd^2:	40

$$\therefore \rho = 1 - \frac{6\Sigma d^2}{n(n^2-1)} = 1 - \frac{6 \times 40}{10 \cdot 99} = 0.76.$$

There is a good relation between Mathematics & Statistics.

Prob: Ten Competetors in a musical test were ranked by the 3 Judges A, B and C in the following order.

Ranks by A	1	6	5	10	3	2	4	9	7	8
Ranks by B	3	5	8	4	7	10	2	1	6	9
Ranks by C	6	4	9	8	1	2	3	10	5	7.

Using the rank correlation Method, discuss which pair of Judges has the nearest approach to the common liking in music

Sol:- here N = 10

Ranks By A (X)	Ranks By B (Y)	Ranks By C (Z)	D_1 X−Y	D_2 X−Z	D_3 Y−Z	D_1^2	D_2^2	D_3^2
1	3	6	−2	−5	−3	4	25	9
6	5	4	1	2	1	1	4	1
5	8	9	−3	−4	−1	9	16	1
10	4	8	6	2	−4	36	4	16
3	7	1	−4	2	6	16	4	36
2	10	2	−8	0	8	64	0	64
4	2	3	2	1	−1	4	1	1
9	1	10	8	−1	−9	64	1	81
7	6	5	1	2	1	1	4	1
8	9	7	−1	1	2	1	1	4

$\Sigma D_1^2 = 200 \quad \Sigma D_2^2 = 60 \quad \Sigma D_3^2 = 214$

$\therefore P_1 ($

$$\rho_1(X,Y) = 1 - \frac{6\Sigma D_1^2}{n(n^2-1)} = 1 - \frac{6\times 200}{10\times 99} = \frac{-7}{33}$$

$$\rho_2(X,Z) = 1 - \frac{6\Sigma D_2^2}{n(n^2-1)} = 1 - \frac{6\times 60}{10\times 99} = 7/11$$

$$\rho_3(Y,Z) = 1 - \frac{6\Sigma D_3^2}{n(n^2-1)} = 1 - \frac{6\times 214}{10\times 99} = \frac{-49}{165}$$

Since $\rho_2(X,Z)$ is maximum (+ve)

We conclude that the pair of judges A and C has here nearest approach to common linking in music.

Equal or Repeated Ranks:

If any two or more persons are bracketed equal in any Classification or if there is more than one item with the same value in the series then the spearman's Formula for Calculating the rank correlation coefficient break down. In this case common ranks are given to repeated items. The common rank is the average of the ranks which these items would have assumed, if they were different from each other and the next item will get the rank next to ranks allready assumed.

For example: If two individuals are placed in the 7th place, each of them are given the rank 7-5 and the next rank will be 9. Ily if 3 are ranked equal to the 7th place then they are given by rank $\frac{7+8+9}{3} = 8$. which is the common rank assigned to each, and the next rank will be 10.

We use a slightly different formula.

$$\rho = 1 - 6\left\{\frac{\sum D^2 + \frac{1}{12}(m^3-m) + \frac{1}{12}(m^3-m)\ldots}{N(N^2-1)}\right\}$$

Where m is the number of items whose ranks are common.

Prob: From the following data calculate the rank correlation coefficient after making adjustment for tied ranks.

X	48	33	40	9	16	16	65	24	16	57
Y	13	13	24	6	15	4	20	9	6	19

Sol. First we have to assign ranks of the variables.

X	Rank x	Y	Rank y.	$D = x - y$.	D^2
48	3	13	5.5	−2.5	6.25
33	5	13	5.5	0.5	0.25
40	4	24	1	3	9
9	10	6	8.5	1.5	2.25
16	8	15	4	4	16
16	8	4	10	2	4
65	1	20	2	1	1
24	6	9	7	1	1
16	8	6	8.5	0.5	0.25
57	2	19	3	1	1.00

$\Sigma D^2 = 41$

16 is repeated 3 times X items hence $m=3$

and 13 and 6 repeated twice in Y $m=2$.

$$\therefore \rho = 1 - \frac{6\left[\Sigma D^2 + \frac{1}{12}(m^3-m) + \frac{1}{12}(m^3-m) + \frac{1}{12}(m^3-m)\right]}{N(N^2-1)}$$

$$= 1 - 6\left[\frac{41 + \frac{1}{12}(3^3-3) + \frac{1}{12}(2^3-2) + \frac{1}{12}(2^3-2)}{990}\right]$$

$$= 0.733.$$

Regression.

The study of correlation measures the direction and strength of the relationship between two variables.

In correlation we can estimate the value of one variable when the value of the other variable is given. Since price and supply correlated, we can find out the expected amount of supply for a given price or we can find out the price to get a given amount of supply.

In regression, we can estimate value of one variable with the value of the other variable which is known. The statistical method which helps us to estimate unknown value of the variable from the known value of the related variable is called regression. The line described as the average relationship between two variables is known as line regression.

Now a days we using the terms estimating line of regression line.

Uses:

1. It is ~~esti~~ used to estimate the relation between two economic variables like Income and Expenditure
2. It is a highly valuable tool in Economics and Business
3. It is widely used in prediction purpose.
4. We can calculate coefficient of correlation

5. It is usefull in statistical estimation of demand curves, Supply curves, products on function, cost function and consumption function etc.

Comparison between correlation and Regression

The correlation coefficient is a measure of degree of covariability between two variables, while the regression establishes a functional relation between dependent and independent variables, so that future can be predicted for a given value of the later. In correlation both the variable x and y are random variables, where as in Regression, x is a random variable and y is fixed variable.

The coefficient of correlation is a relative measure where as Regression coefficient is an absolute figure.

Find the regression line using least squares Method of curve fitting.

Regression Equation:

Regression equation is a algebraic expression of the regression line. It can be classified into Regression eqns, Reg. coeff, Individual observation and group discussion.

The standard Form of the Regression equation $Y = a + bx$ where a, b are called Constants a indicates the value of Y when $x = 0$. It is called Y-intercept. b indicates the value of slope of the regression line and gives the measure of change of Y for unit change in X.

It is also called the Regression coefficient of Y on X.

If the equation of the line is $X = a + bY$.

then the constants a ~~and it~~ indicates the X- intercept and b indicates the slope of the regression line and is known as Regression coefficient of X on Y.

Regression Equation Y on X.

Normal Equations are

$$\sum Y = Na + b\sum X$$

$$\sum XY = a\sum X + b\sum X^2.$$

Regression Eqn. X on Y:

Normal Equations are

$$\sum X = Na + b\sum Y$$

$$\sum XY = a\sum Y + b\sum Y^2.$$

Deviation taken from Arthemetic means of X on Y.

This method is easier and simpler then the previous method to find the values of a and b.

We can find out the deviation of X and Y series from their respective means.

Regression equation of X on Y is given by

$$X - \bar{X} = r\frac{\sigma_x}{\sigma_y}(Y - \bar{Y})$$

$\bar{x}$ = mean of $\bar{X}$ series

$\bar{Y}$ " $\bar{Y}$ "

The regression coefficient of X on Y $= b_{xy} = \frac{r\sigma_x}{\sigma_y}$

$$= \frac{\Sigma xy}{\Sigma y^2} \quad x = X - \bar{X}, \; y = Y - \bar{Y}$$

The regression coefficient of Y on X $= b_{yx} = \frac{\sigma_y}{\sigma_x}$

$$= \frac{\Sigma xy}{\Sigma x^2} \quad x = X - \bar{X} \quad y = Y -$$

Thus $r^2 = b_{xy} \times b_{yx}$.

Result: $X - \bar{X} = r\frac{\sigma_x}{\sigma_y}(Y - \bar{Y})$.

r = correlation coefficient

σ = S.D.

Prob: For the following data, find equation of two regression lines.

X	1	2	3	4	5
Y	15	25	35	45	55

Calculation of regression coefficients.

X	Y	$x = X - \bar{X}$	$y = Y - \bar{Y}$	y^2	x^2	xy
1	15	−2	−20	400	4	40
2	25	−1	−10	100	1	10
3	35	0	0	0	0	0
4	45	1	10	100	1	10
5	55	2	20	400	4	40
15	175			1000	10	100

$N = 5$

$\bar{X} = \frac{15}{5} = 3$

$\bar{Y} = \frac{175}{5} = 35$

Now $b_{yx} = \frac{\sum xy}{\sum x^2} = \frac{100}{10} = 10.$

∴ Regression equation of y on x is given by

$$y - \bar{y} = b_{yx}(x - \bar{x})$$

$$y - 35 = 10(x - 3)$$

Now Regression equation of x on y is given by

$$x - \bar{x} = b_{xy}(y - \bar{y}). \qquad b_{xy} = \frac{\sum xy}{\sum y^2} = \frac{100}{1000} = \frac{1}{10}$$

$$x - 3 = \frac{1}{10}(y - 35).$$

Prob: In the following table S is weight of potassium bromide which will dissolve in 100 gms. of water at V°C. Fit an equation of the form $S = mT + b$ by the method of least squares. Use this relation to estimate S when $T = 50°$.

T	0	20	40	60	80
S	54	65	75	85	96.

T	S	$d_T = \frac{T-40}{N}$	d_T^2	$d_S = \frac{T-75}{10}$	d_S^2	$d_T d_S$
0	54	−4	16	−2.1	4.41	8.4
20	65	−2	4	−1.0	1.00	2.0
40	75	0	0	0	0.00	3.0
60	85	2	4	1.0	1.00	2.0
80	96	4	16.	2.1	4.41	8.4
$\Sigma T = 200$	375			0.	10.82	20.8

$N = 5$

Avg = 40.

100 grin
1 kg = $\frac{1}{10}$.

$$m = \frac{\Sigma d_T d_S}{\Sigma d_T^2} = \frac{20.8}{40} = 0.52.$$

and b is given by the equation

$$\Sigma S = m \Sigma T + nb.$$

$$\therefore 375 = (0.52)(200) + 5b \Rightarrow b = 54.2$$

When $T = 50°C$ $\quad S = 0.52 \times 50 + 54.2$

$= 80.2$

When $T = 50°C$ $\quad S = 0.52 \times 50 + 54.2$

$= 80.2$

EXERCISE

1. Calculate the coefficient of correlation between X and Y from the following data :

X:	1	2	3	4	5	6	7
Y:	2	4	5	3	8	6	7

2. The following data gives the marks obtained by 10 students in accountancy and statistics.

R. No.	Accountancy	Statistics
1	45	35
2	70	90
3	65	70
4	30	40
5	90	95
6	40	40
7	50	80
8	75	80
9	85	80
10	60	50

Find the coefficient of correlation and interpret it.

3. Calculate coefficient of correlation between age and playing habit from the following data.

Age in years	Population	No. of players
15 - 20	1500	1200
20 - 25	2000	1560
25 - 30	4000	2280
30 - 35	3000	1500
35 - 40	2500	1000
40 - 45	1000	300
45 - 50	800	200
50 - 55	500	50
55 - 60	200	6

4. The following table gives the distribution of the total population and those who are totally and partially blind among them. Find the coefficient of correlation.

Age	No. of Persons	Blind
0 - 10	100	55
10 - 20	60	40
20 -30	40	40
30 - 40	36	40
40 - 50	24	36
50 - 60	11	22
60 - 70	6	18
70 - 80	3	15

5. The following table gives the scores obtained by 11 students in English and Telugu Translation. Find the rank correlation coefficient.

Scores in English	40	46	54	60	70	80	82	85	85	90	95
Scores in Telugu	45	45	50	43	40	75	55	72	65	42	70

6. The ranks of the 15 students in two subjects A and B are given below, the two numbers within the brackets denoting the ranks of the same student in A and B respectively.

(1,10), (2,7), (3,2), (4,6), (5,4), (6,8), (7,3), (8,1), (9,11), (10,15), (11,9), (12,5), (13,14), (14,12), (15,13)

Use Spearman's formula to find the rank correlation coefficient.

ANSWERS

1. $r = 0.79$
2. $r = 0.903$
3. $r = -0.993$
4. $r = 0.898$
5. $r_5 = 0.359$
6. $r_5 = 0.514$

EXERCISE

1. Given the bi-variate data :

X :	1	5	3	2	1	1	7	3
Y :	6	1	0	0	1	2	1	5

(*a*) Find the Regression line of Y on X and hence predict Y if X = 10.

(*b*) Fit a Regression line of X on Y and hence predict X if Y = 2.5.

(*c*) Calculate Karl Pearson's correlation coefficient.

2. Calculate the correlation coefficient and regression coefficient for the following data.

X :	2	4	6	8	10	12	14
Y :	4	2	5	10	4	11	12

Find the estimate of y when $x = 13$

3. Find the most likely production corresponding to a rainfall 40" from the following data :

	Rainfall	Production
Average	30"	500 kgs
Standard deviation	5"	100 kgs
Coefficient of Correlation	0.8	

4. Heights of fathers and sons are given in inches :

Height of father	65	66	67	67	68	69	71	73
Height of son	67	68	64	68	72	70	69	70

Form the two lines of regression and calculate the expected average height of the son when the height of the father is 67.5 inches.

ANSWERS

1. (*a*) $Y = 2.874 - 0.304\,X$, when $X = 10, Y = -0.166$

(*b*) $X = 3.431 - 0.278\,Y$, when $Y = 2.5, X = 2.736$

(*c*) $r = -0.29$

2. Regression coefficient of X on Y $= 0.847$,

Regression coefficient of Y on X $= 0.732$

Coefficient of correlation $= 0.787$

Correlation equation is $Y = 0.732\,X + 1.001$

when $X = 13, Y = 10.571$

3. The most likely production corresponding to a rainfall of 40" is 660 kgs.

4. $Y = 0.421X + 39.77$; $X = 0.524Y + 32.9$

when $X = 67.5", Y = 68.19"$